PALEOMAGNETISM OF THE ALLARD LAKE ANORTHOSITE SUITE

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Analysis of paleomagnetic data on 112 samples of Precambrian anorthosite, norite, pyroxene–syenite and hemo-ilmenite ore from four main localities in the Allard Lake area reveals: (1) A relatively soft magnetic component resides in magnetite, which is of variable abundance in all rock types. The magnetite remanent magnetic vector is of normal polarity, and is more or less directly opposed to a hard component residing in hemo-ilmenite, the predominant oxide. Self-reversal of the permanent magnetism in the hemo-ilmenite is the favored explanation. (2) Disregarding polarity, the stable remanent magnetic vector has similar orientation in all rock types from all four localities ($N = 4$, $\alpha_{95} = 18.0$). This precludes significant differential rotation of individual structural blocks since the rocks became magnetized. Furthermore, either no polar wander occurred during the emplacement and cooling of the entire igneous rock suite, or all these rock types acquired their magnetism simultaneously during subsequent metamorphism. (3) On the centered-axial-dipole model for the earth's magnetic field, the remanent vector in the Allard Lake rocks gives a pole southeast of Newfoundland, at coordinates 38°36′ N., 39°36′ W. ($\alpha_{95} = 18.0$).

INTRODUCTION

Previous study of the rock magnetism of the Allard Lake ilmenite deposits by Carmichael (1958, 1961, 1964) and Hargraves (1959) has established the following relationships.

(1) The permanent magnetism of the ilmenite[3] ore resides in the exsolved titanhematite lamellae (Carmichael 1959).

(2) The ore shows a variable degree of fabric anisotropy, manifest by a preferred planar orientation of the basal-planes of the rhombohedral ilmenite crystals. This fabric anisotropy causes a marked magnetic susceptibility anisotropy. The permanent magnetic vector of the sample lies in or near this plane of maximum susceptibility, and its magnitude is a function of the orientation of this plane with respect to an inferred magnetizing field (Hargraves 1959).

(3) The reverse polarization of the hemo-ilmenite ore (i.e. North-seeking pole inclined upwards) is concluded to have arisen by an order–disorder self-reversing mechanism, even though the process cannot be duplicated in the laboratory (Carmichael 1961).

(4) Evidence of oppositely polarized magnetic minerals (magnetite and hemo-ilmenite) coexisting in the same rock was reported by Hargraves (1959, p. 1577) and by Carmichael (1964).

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[3]The ilmenite occurring in the ore deposits and rocks of the Allard Lake area is actually a micro-intergrowth of ilmenite and hematite, i.e. hemo-ilmenite or ilmeno-hematite. The general term 'ilmenite' is, however, frequently used in this paper, and unless specifically stated otherwise it should be understood to signify the intergrowth.

Despite this intensive study, essentially no paleomagnetic data have resulted. Hargraves (1959) obtained a field vector for the Lac Tio ilmenite deposit, but not knowing whether the Lac Tio deposit had remained undisturbed since its magnetization, he did not attempt the calculation of a "pole position". The directions of natural remanent magnetism (n.r.m.) of other rock samples collected throughout the Allard Lake area were widely scattered, and for lack of facilities for magnetic cleaning in 1959, the paleomagnetic study was not concluded.

With the equipping of Princeton University's Rock Magnetism Research Laboratory the investigation has been resumed. Both a-c. and thermal demagnetization of the rocks have proved the existence of a stable magnetic component with relatively uniform orientation in all rock types throughout the area. These range in mineralogy from anorthosite, through hemo-ilmenite ore, to pyroxene syenite (Hargraves 1962). The consistency of the results from samples collected throughout an area of 40 square miles suggests that there has been no substantial differential rotation of structural blocks since the rocks of this area last acquired a permanent magnetization.

Assuming a centered axial dipole model for the earth's magnetic field (Irving 1964), the average orientation of the permanent magnetism of the Allard Lake rocks points to a pole located southeast of Newfoundland at coordinates 38.6° N., 39.6° W.

GENERAL GEOLOGY

The general geology and petrology of the Allard Lake area (51° N., 63° W.) has been described by Hargraves (1962). On the basis of field and petrographic evidence the paragenetic sequence established, from oldest to youngest, was: anorthosite; oxide-rich norite; hemo-ilmenite ore deposits; and syenite. The emplacement of these rocks was considered to constitute essentially one continuous petrogenetic cycle, the absolute duration of which was, however, unknown.

SAMPLING AND MEASUREMENT

The 112 oriented samples utilized in this study were collected mainly during the course of geological mapping, rather than with paleomagnetic study primarily in mind. As a result, sample coverage is relatively broad and even, rather than concentrated at numerous sites. Nevertheless, the samples can be separated reasonably into four principal localities, named as follows: (1) Lac Tio Mine—48 samples; (2) Grader Railroad—13 samples; (3) Lac Allard-McRae Ridge—21 samples; and (4) Lac Puyjalon—21 samples. In addition, three ilmenite samples from the Northwest Arm ilmenite deposit were studied, and six from around Lac Ellen.

Because of the strong magnetic anomalies associated with the ilmenite deposits and the oxide-rich norite sheets, an azimuthal orientation of all samples was made with reference to some prominent landmark (or survey beacons at Lac Tio Mine). The actual orientation of the azimuthal line was then obtained from aerial photographs. A Brunton compass was used only to

mark the horizontal plane on the samples. The accuracy of orientation of individual samples is estimated to be $\pm 5°$.

At different times, r.m. measurements were made on an astatic magnetometer (McElhinny 1963) and a spinner magnetometer (Phillips 1965; Phillips and Kuckes 1966). As the intensity of magnetization of these rocks was generally greater than 10^{-4} e.m.u./cc the experimental error is estimated to be less than 2°.

The a-c. demagnetization equipment used has been described by McElhinny (1963) and that for thermal demagnetization by Phillips (1964). The values given for the magnitude of the a-c. demagnetizing fields are peak values, not r.m.s.

MAGNETISM OF ROCK TYPES

Anorthosite

Samples containing in excess of 85% plagioclase (An_{40-50}) were classified as anorthosite. Those anorthosite samples with between 5 and 15% ferromagnesians were qualified as noritic or gabbroic anorthosite. The usual varietal

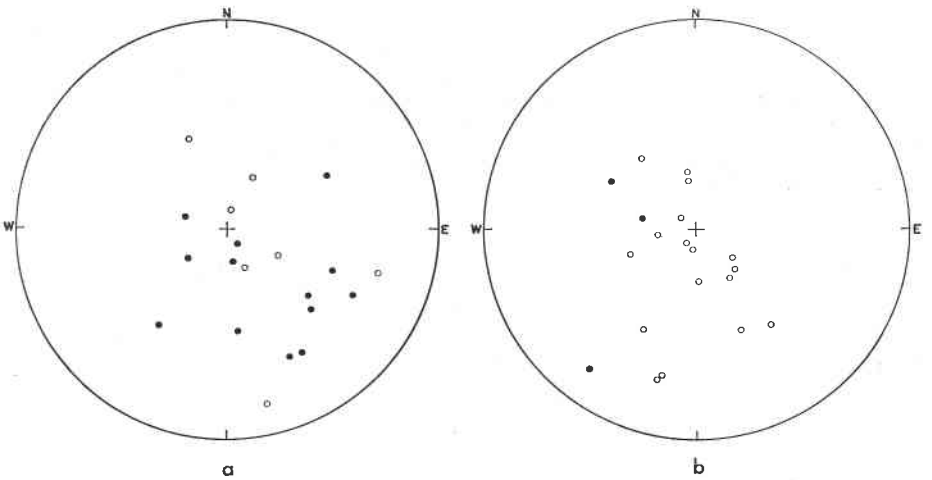


FIG. 1. Remanent magnetism in anorthosite: (*a*) n.r.m. and (*b*) after thermal demagnetization to 600 °C. Equal-area projection; dots (open-circles) signify north-seeking pole plotted on lower (upper) hemisphere.

pyroxene is bronzite (Mg_{72}).[4] Although the n.r.m. of anorthosite specimens from the same sample were usually within 10°, the n.r.m. of samples from the same site were widely scattered and of mixed polarity (Fig. 1).

Those specimens with reverse n.r.m. changed little in intensity or orientation on demagnetization (to peak a-c. fields of 1000 Oe or 600 °C), whereas with three exceptions those normal to start with reversed on demagnetization. An a-c. field of 500 Oe was usually sufficient to cause the reversal; however, with thermal demagnetization, the change invariably occurred between 550 °C and

[4]Low-grade ore zones at Lac Tio Mine, consisting of a banded admixture of plagioclase and hemo-ilmenite, are designated as foliated anorthosite in Table IV.

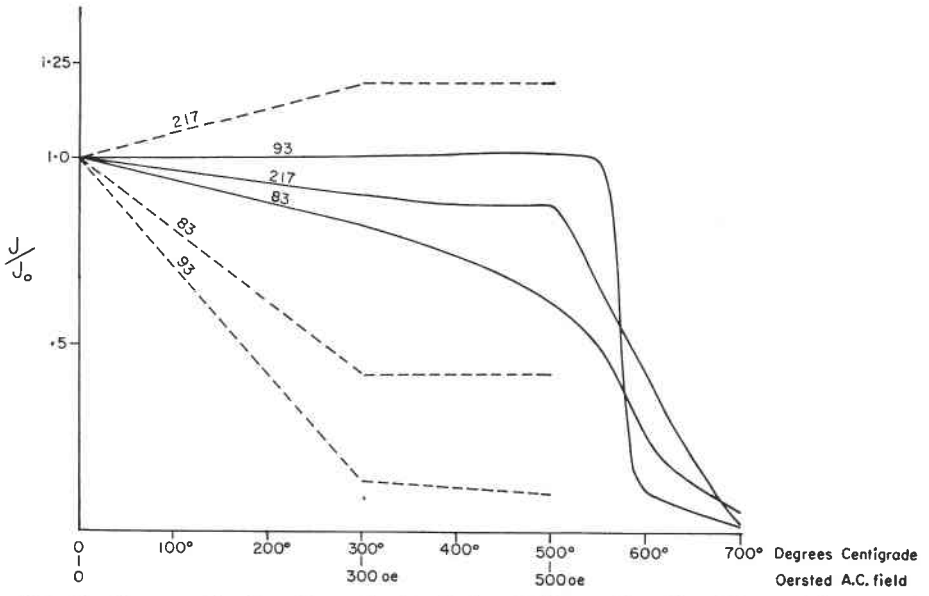


FIG. 2. Demagnetization of anorthosite. Ratio of J/J_0 as a function of thermal demagnetization (solid lines) and a-c. demagnetization (broken lines) for separate specimens from three samples. Specimens from Samples 83 and 93 had normal n.r.m., which reversed both on a-c. demagnetization, and on thermal demagnetization between 550 and 600 °C; sample 217 had reverse n.r.m., and changed little in orientation on demagnetization. Results suggest significant magnetite content in 83 and 93, negligible in 217.

600 °C (see Fig. 2). The remanent-magnetism orientations before and after demagnetization are shown in Fig. 1, the mean for Lac Tio anorthosites in Table IV and Fig. 5.

Ilmeno-hematite is the characteristic, but very minor, oxide accessory seen in polished sections of 'pure' anorthosite. The very small, opaque, oriented rod-like inclusions in the plagioclase crystals also seem to be of this mineral. Magnetite was seen in only one of eight polished sections, where it seemed to be confined to the contact between pink and grey anorthosite zones. Nevertheless, the thermal demagnetization curves strongly suggest the presence of magnetite of magnetic polarity opposed to that of the ilmeno-hematite.

Oxide-rich Norite

The essential constituents of the norite, in paragenetic order, are plagioclase (An_{40-45}), pyroxene (orthorhombic Mg_{60-65}, and monoclinic), apatite, and oxide, the mafic constituents generally totalling in excess of 50% by weight. A gneissosity, evident in outcrop, is rarely discernible in thin section.

The total oxide content, measured in 14 samples of norite, varied from 20 to 50%, averaging about 30%, by weight. The proportion of ilmenite to magnetite varies from approximately equal to predominantly hemo-ilmenite.

Previous work (Hargraves 1959, p. 1577; Carmichael 1964) has demonstrated that the magnetite and ilmenite coexisting in the norite have opposite magnetic polarities and that the normal component residing in the magnetite can be

preferentially erased by a-c. demagnetization. Again, in this study, on demagnetization all normal samples reversed, the magnetization vector in some samples migrating through nearly 180° (see Fig. 3). Some samples, normal to start with, were completely reversed after 300 Oe a-c. demagnetization. Two

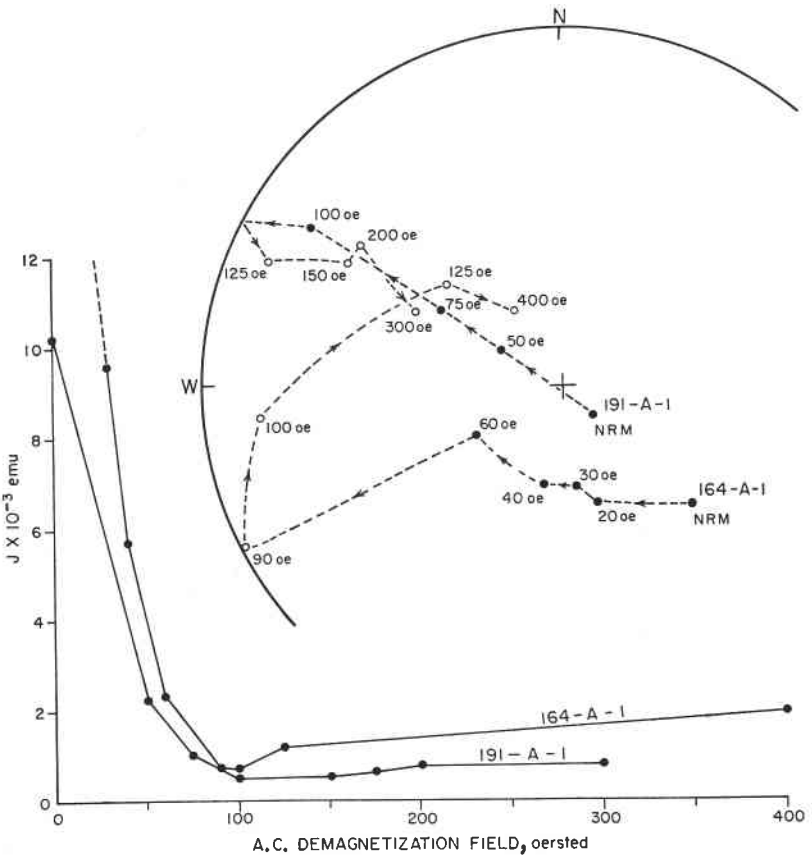


FIG. 3. Variation in r.m. vector in oxide-rich norite on systematic a-c. demagnetization. Note rapid migration from normal to reverse between 75 Oe and 150 Oe, and minimum intensity around 100 Oe.

samples were still migrating after 1000 Oe (the maximum a-c. field available on our equipment).

Because of the oxidation of the magnetite that occurs on thermal demagnetization in air at high temperature, this technique was not applied systematically to the oxide-rich norites.[5]

[5]Two samples that had not completely reversed after 1000 gauss a-c. fields, however, were thermally demagnetized at temperatures up to 600 °C. Surprisingly, they became strongly positive after this treatment. A-c. fields of 1000 Oe again succeeded in partly reversing them, but after thermal demagnetization they again became positive. These results are thought to indicate magnetostatic interaction: on cooling from 600 °C in field-free space, the magnetite acquires a t.r.m. in the demagnetizing field of the hemo-ilmenite. Further investigation of this behaviour is in progress.

The average r.m. vectors of norite samples from the different localities, after a-c. demagnetization, are summarized in Table IV, and projected in Fig. 5.

Hemo-ilmenite Ore

The mineralogy and magnetic properties of Allard Lake hemo-ilmenite ore have been investigated previously in considerable detail (Carmichael 1959, 1961; Hargraves 1959, 1962; Uyeda *et al.* 1963). A summary of this work follows.

The ore averages in excess of 85% hemo-ilmenite, with plagioclase the principal gangue. Pyrite and (or) pyrrhotite are widely distributed in very minor amounts. Magnetite is rare. The individual hemo-ilmenite grains show marked susceptibility anisotropy (Uyeda *et al.* 1963), with a maximum in the basal plane and a minimum along the C-axis. A preferred crystallographic orientation of the grains in the ore imparts to it a similar anisotropy of varying intensity (Hargraves 1959). The permanent magnetism of the ore lies in or near the plane of maximum susceptibility (Hargraves 1959, p.1572). On the assumption that the observed n.r.m. vector represented a component of the magnetizing field, which had been resolved to or towards the plane of maximum susceptibility, a magnetizing field direction was obtained by graphical means. The validity of the assumption was confirmed by the correlation between the intensity of n.r.m. of samples and the angle between the magnetizing field and the plane of maximum susceptibility.

In this study, it was found that a-c. demagnetization had little effect on most ilmenite samples. Any change that did occur was completed in fields of 300 Oe and generally resulted in a slight *increase* in intensity. As the n.r.m. of the ilmenites was reversed to begin with, this increase in intensity was considered to indicate the presence of a soft normal component, which is erased on demagnetization. This could be either the primary n.r.m. of traces of magnetite (or pyrrhotite) or a viscous component acquired in the present field. The orientation and intensity of r.m. remained unchanged between 300 and 1000 Oe.

Using the refined remanent magnetism data, and the original susceptibility anisotropy data, calculation of the magnetizing field orientation was done algebraically and with a computer, rather than graphically, as in 1959. The cross product of the permanent magnetism vector ($\boldsymbol{jr}$) and the vector representing the direction of minimum susceptibility closest to the r.m. vector ($\boldsymbol{k}_{\min}$) defines the pole ($\boldsymbol{p}$) to the great circle plane along which the permanent magnetism vector 'migrated': $\boldsymbol{p} = \boldsymbol{jr} \times \boldsymbol{k}_{\min}$. This 'migration' consisted of a rotation (or resolution) away from the magnetizing field toward the plane of maximum susceptibility in the specimen. As such, the rotation plane includes the magnetizing field vector. Every intersection of two such planes is a unique solution to the magnetizing field vector $\boldsymbol{H}$. The cross product of poles to two such planes gives this solution (see Fig. 4):

$$\boldsymbol{H} = \boldsymbol{p}_1 \times \boldsymbol{p}_2 = (\boldsymbol{jr}_1 \times \boldsymbol{k}_{\min 1}) \times (\boldsymbol{jr}_2 \times \boldsymbol{k}_{\min 2})$$

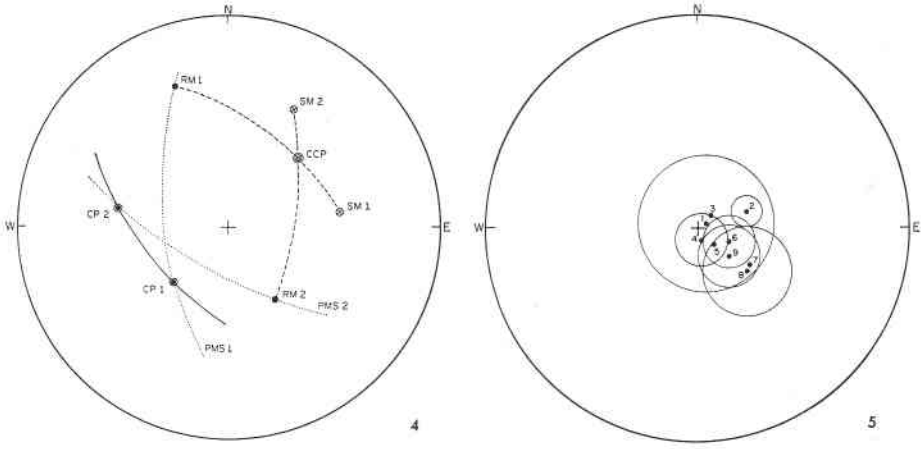


FIG. 4. Stereographic illustration of cross-product technique for obtaining orientation of magnetizing field in hemo-ilmenite ore. PMS, plane of maximum susceptibility; SM, susceptibility minimum (=pole to PMS); RM, remanent magnetism; CP, pole to SM–RM plane (CP = SM × RM); CCP, intersection of two (SM–RM) planes = inferred magnetizing field direction = CP1 × CP2.

FIG. 5. Equal-area, *lower hemisphere*, projection of mean r.m. vectors from different rock-types and localities. Data in Table V. 1, 2, and 3, Lac Tio anorthosite, foliated anorthosite, and hemo-ilmenite cross-product vector; 4 and 5, McRae oxide-rich norite, and hemo-ilmenite cross-product vector; 6, Grader oxide-rich norite; 7, 8, and 9, Puyjalon anorthosites, oxide-rich norites, and syenites (the reverse, normal-reverse, and normal groups). The confidence cones approximate circles near the center of equal area projections, and have been so drawn.

A program was written to compute (*a*) the lower hemisphere vector, and (*b*) the vectors lying in the hemisphere whose center plunges 45° E., for every possible combination of specimen pairs, and to calculate their mean and resultant. The results are presented and compared with the simple vector mean of the r.m. of all the individual specimens in Table I.

Syenites

The contact between the more mafic gneissose anorthosite of the border facies and the surrounding pyroxene syenite was difficult to locate in the field. Where best exposed along the railroad track, it was found to be gradational (Hargraves 1962, p. 180). The samples oriented for paleomagnetic study were

TABLE I

Locality		Cross-product					R.M. vector mean				
		Specimens	Intersections	D	I	*k*	Samples	D	I	*k*	*α95*
Lac Tio	(*a*)	49	1176	47.9	83.3	6.6	26	95.4	62.3	3.4	18.2
	(*b*)			51.3	81.0	6.6					
Lac Allard–McRae	(*a*)	21	210	136.6	81.7	6.2	11	139.3	65.5	11.4	14.1
	(*b*)			126.0	75.1	5.9					

NOTE: Allard Lake hemo-ilmenites. Comparison of simple r.m. vector means, and the vector obtained by the double cross-product technique described in the text. In computing the latter, if the scatter of intersections closest to the true mean should include some on the upper hemisphere, the programmed computation of the mean *lower hemisphere* vector only (vector *a*), will cause the vector to be steeper than it really is. To check this possibility the reference hemisphere was rotated 45° to the east, and the mean recalculated (vector *b*). Although in both cases the inclination does decrease slightly, the value of *k* also decreases, indicating greater scatter. The mean *lower hemisphere* vector is favored, and has been used in all subsequent computations.

collected north and east of Lake Puyjalon. In the contact zone, the correct designation of a particular outcrop as "syenite" or "anorthosite" was difficult, and indeed in many outcrops the rock was more or less a hybrid.

Magnetically, all samples from near Lake Puyjalon could be classified into three groups: (1) reversed n.r.m. (includes most anorthosite samples); (2) normal n.r.m., which became reversed on a-c. or thermal demagnetization; (3) normal n.r.m., unchanged on demagnetization (includes most syenite samples).

The oxide in the anorthosite (group 1) is predominently hemo-ilmenite; that in the syenite is predominantly magnetite. What little ilmenite is found in the syenite shows rare hematite lamellae.

Group 2 includes samples which from their geographic location belong with the syenite. Their intermediate magnetic character is compatible with the petrologic evidence of the transitional relationship between anorthosite and syenite (Hargraves 1962, p. 178).

The lower hemisphere projections of the vector means after demagnetization for groups 1, 2, and 3, are shown in Fig. 5. There is little difference between them.

VECTOR ERASED ON DEMAGNETIZATION

In the rock-magnetism literature, it appears that little consideration has been given to the *orientation* of the 'soft-component' erased on demagnetization, by either a-c. or thermal techniques. The composition of this component could be complex, especially in older rocks, but in rocks of high susceptibility, the viscous magnetization should include a contribution from the ambient magnetic field at the sampling site at the time of sampling. Furthermore, this viscous component will be the least stable on a-c. demagnetization and the magnetic vector erased in the first stage of a systematic demagnetization experiment, should at least *include* this component due to the present earth's field, if not be composed of it entirely. In principle this offers a means of azimuthally orienting drill core from vertical holes in appropriate rock types.

Likewise, if two distinct ferromagnetic minerals of contrasting coercivities are present in the same rock, systematic demagnetization should preferentially erase the component of r.m. due to the mineral of lower coercivity (in a-c. demagnetization) or blocking temperature (in thermal demagnetization).

The experience of this study in systematic demagnetization of oxide-rich norite and syenite is that the component of r.m. residing in magnetite is of opposite polarity to that of coexisting hemo-ilmenite, and can be preferentially erased by a-c. demagnetization up to 1000 Oe, or thermal demagnetization to 600 °C.

The increments of the magnetite vector erased from a given specimen on a-c. demagnetization should, therefore, in principle all be parallel and opposite in direction to the stable hemo-ilmenite vector (if present; see p. 365 of this text). In practice, because of the intrinsic magnetic anisotropy of hemo-ilmenite, and the gneissose fabric of the rocks, this perfection in orientation is not to be expected.

A program was written to compute (*a*) the vector erased by each demagnetization step, (*b*) the total vector erased from the n.r.m. and (*c*) the total vector erased after the first demagnetization step. A sample of the output is presented in Table II.

The results showed that the orientation of the vector erased by each step is consistent as long as its intensity is more than about 5% of the n.r.m. From one sample to another, the total vector erased from n.r.m. is less uniform than that erased after the first step. This can be reasonably attributed to viscous magnetization, which is erased in 25–50 Oe.

The average orientation of the total vector erased after the first step in the demagnetization experiment (usually 25 gauss) is compared with the stable vector remaining, for different localities, in Table III. In addition, is shown the average orientation of the vector erased between 500 and 600 °C on thermal demagnetization of those anorthosite specimens that started with positive inclination and ended reversed.

A statistical test to determine whether in each case, the mean residual (hemo-ilmenite) and the (reversed) erased (magnetite) vectors are different (Watson and Irving 1957, p. 292) was applied to the four examples given in Table III. The two vectors from anorthosite and the Grader oxide-rich norite are not significantly different at the 5% level, those from Puyjalon are just different at the 5% level but not at the 1% level, and those from the McRae samples are different.

The results from Puyjalon and McRae are not entirely consistent with the ideal model of two coexisting oppositely polarized magnetic mineral constituents. It should be pointed out, however, that this test of the model assumes that only the magnetite component is erased on a-c. demagnetization, and that only the reversed hemo-ilmenite vector remains after demagnetization. If fractions of both components are erased by and (or) remain after demagnetization, the test is not valid. The presence of a hard normal component in the Puyjalon syenites, in which hemo-ilmenite is rare or absent, implies that at least some fraction of the magnetite has high coercivity. Likewise some fraction of the hemo-ilmenite may be soft. The authors favor this general explanation for the inconsistencies in the test of the Puyjalon and McRae samples, and consider the weight of evidence in favor of the model of oppositely polarized magnetite and hemo-ilmenite.

SUMMARY OF RESULTS

The magnetic results are summarized in Table IV and Fig. 5.

It is clear that in all rock types in which magnetite and hemo-ilmenite coexist the magnetite is relatively 'soft' and the ilmenite is 'hard'. A-c. fields of 100 to 150 Oe were adequate to demagnetize 90% of the magnetite component of n.r.m., at least in those rocks containing coarse-grained magnetite. The hemo-ilmenite is virtually unaffected in a-c. fields of 1000 Oe.

Vector subtraction of the r.m. after demagnetization from the n.r.m. shows that the polarity of the 'soft' magnetite is normal, that of the hemo-ilmenite

TABLE II

Sample No.	Vector H Oe	Residual D	Residual I	Residual Je.m.u. ×1000	Erased D	Erased I	Erased Je.m.u. ×1000	Cumulative Erased from n.r.m. D	Cumulative Erased from n.r.m. I	Cumulative Erased from n.r.m. Je.m.u. ×1000	Cumulative Erased from 20 Oe D	Cumulative Erased from 20 Oe I	Cumulative Erased from 20 Oe Je.m.u. ×1000
$164a_1$	N	134.0	49.9	44.2									
	20	164.0	62.7	14.9	125.4	42.0	30.7	125.4	42.0	30.7			
	30	173.7	66.5	9.74	151.7	54.5	5.29	128.4	44.2	35.7	151.7	54.5	5.29
	40	191.8	66.6	5.70	151.6	63.2	4.15	130.0	46.3	39.5	151.7	58.3	9.41
	60	240.0	67.0	2.34	169.4	59.8	3.56	132.2	47.8	42.8	156.4	59.0	12.9
	90	242.8	− 1.6	0.74	228.4	85.3	2.18	132.6	49.7	44.4	157.8	63.1	14.9
	100	264.3	−16.4	0.75	167.7	35.0	0.33	132.9	49.6	44.7	158.2	62.5	15.2
	125	312.0	−54.4	1.19	200.6	52.9	0.95	134.0	50.0	45.4	161.2	62.4	16.0
	400	327.5	−69.5	1.95	232.8	77.8	0.88	134.3	50.7	46.0	162.5	63.6	16.8

NOTE: Vector erased by systematic a-c. demagnetization of a specimen of oxide-rich norite, from railroad in the Grader locality. An example of the computer program output.

TABLE III

Comparison of average residual vector with average vector erased (*a*) between 500 and 600 °C in thermal demagnetization of *anorthosite*, and (*b*) between 25 and 500 Oe in a-c. demagnetization of specimens from Grader, McRae, and Puyjalon.

	Specimens	N_1 = samples	Residual vector R_1	D_1	I_1	k_1	$\alpha 95$	Specimens	N_2 samples	Erased vector (Reversed) R_2	D_2	I_2	k_2	$\alpha 95$	Combined Sum	R	k	Test $\frac{N\text{-}2\,(\Sigma Ri - R)}{N - \Sigma Ri}$	$F_{2,2(N-2)}$ 5%	1%
Anorthosite	12	12	9.49	180.4	−74.5	4.4	23.5	12	12	10.86	249.0	−83.8	9.6	14.7	24	20.19	6.0	0.97	3.21	5.13
McRae oxide-rich norite	6	5	4.86	25.9	−83.5	28.9	14.5	5	5	4.80	249.0	−58.5	20.5	17.3	10	9.18	11.0	11.7	3.63	6.23
Grader oxide-rich norite	13	8	7.66	296.9	−74.5	20.8	12.4	8	8	6.79	326.5	−75.0	5.8	25.3	16	14.42	9.5	0.3	3.34	5.45
Puyjalon norites and syenites	15	15	13.89	310.7	−73.6	12.7	11.2	15	15	14.17	229.4	−86.6	16.8	9.6	30	27.78	13.1	3.5	3.17	5.01

TABLE IV

Summary of Allard Lake paleomagnetic data.

Locality	Rock type	Demag.*	Spec's.	Samples	D	I	k	$\alpha 95$	Mean D	Mean I	Virtual poles: Pole Latitude	Virtual poles: Pole Longitude	Virtual poles: δp	Virtual poles: δm	Mean pole: Pole Latitude	Mean pole: Pole Longitude	Mean pole: $\alpha 95$
Lac Tio	Anorthosite	Th.	19	11	239.6	−86.4	4.0	26.2									
	Foliated anorthosite	A.C.	12	10	251.0	−70.3	78.9	5.5	244.6	−80.1	+54.9	327.7	24.6	25.7			
	Hemo-ilmenite	A.C. ×-prod.	49		227.9	−83.3	6.6										
	Hemo-ilmenite	A.C.	49	26	275.4	−62.3	3.4	18.2									
Lac Allard–McRae	Oxide-rich norite	A.C.	15	10	346.2	−85.6	27.5	9.4	326.9	−83.8	+39.9	305.2	21.1	21.4			
	Hemo-ilmenite	A.C. ×-prod.	21		316.6	−81.7	6.2										
	Hemo-ilmenite	A.C.	21	11	319.3	−65.5	11.4	14.1							38.6	320.4	18.0
Grader	Oxide-rich norite	A.C.	15	10	295.7	−77.2	23.6	10.1	295.7	−77.2	+36.0	324.0	17.6	18.9			
	Hemo-ilmenite	A.C. ×-prod.	5		90.1	−78.2	75.6										
	Hemo-ilmenite	A.C.	5	3	220.5	−65.8	8.1	46.6									
Puyjalon	Anorthosite	A.C.	5	5	305.9	−65.7	5.3	36.7									
	Anorthosite–syenite	A.C.	5	5	311.5	−64.9	19.1	17.9									
	Syenite	A.C.	11	11	133.9	+74.1	12.1	13.7									
	All together		21	21	131.0	+70.0	10.8	10.1	311.0	−70.0	+22.2	325.2	15.0	17.4			
Lac Ellen	Oxide-rich norite	A.C.	4	2	240.9	−19.7	0.8	—									
	Hemo-ilmenite	A.C.	6	4	283.7	−48.2	5.1	45.0									
	Hemo-ilmenite	A.C. ×-prod.	6		2.9	−74.6	6.9										
N.W. Arm	Hemo-ilmenite	A.C.	6	3	358.5	−58.2	3.5	79.7									
	Hemo-ilmenite	A.C. ×-prod.	6		147.5	−54.2	3.5										

*Vectors reported are those remaining after demagnetization: Th, thermal; A.C., alternating current.

is reversed, and that these two magnetic components are more or less diametrically opposed. The correlation between polarity and mineralogy (the proportion of magnetite to hemo-ilmenite) in all rock types favors self-reversal, rather than a fortuitous field-reversal for the reversed magnetization of the hemo-ilmenite.

The stable permanent magnetism in all rock types can be related to a single magnetizing field direction. Although differential rotation of independent structural blocks may contribute to the scatter observed in Fig. 5, the directions are not significantly different. This uniformity of magnetization in all rock types of the anorthosite suite, regardless of their position in the paragenetic sequence, allows of one of the following explanations.

1. The magnetization was imposed contemporaneously as a result of heating associated either with the emplacement of the syenite (the last in the paragenetic sequence) or subsequent regional metamorphism.

2. The temperatures never dropped below the Curie points during the entire emplacement history of the anorthosite suite.

3. If considerable time elapsed and cooling occurred between the emplacement of any members of the anorthosite suite, at least no substantial polar wandering occurred during the entire interval.

The average of the poles calculated from the mean r.m. vector of all rock types at each of the four principal localities lies southeast of Newfoundland, at coordinates 38°36′ N., 39°36′ W. (α_{95} = 18.0°).

Two K–Ar age determinations by the Geological Survey of Canada (Leech *et al.* 1963; samples GSC 62–142, 143) average around 1 000 m.y. This reflects the 'Grenville' metamorphism (Gilluly 1965), at which general time the Allard Lake rocks are presumed to have acquired their permanent magnetism.

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